Science: Forces

Skill: The Because Chain (cause, effect, consequence) | Warm-up 5, drill 10, paper 20 min

Writing target

- Every explanation is a chain. Write full sentences where each one connects to the next. A list of facts stays at Level 1.
- Link words: so / which means / because / therefore / as a result / this causes / until

The Because Chain

- Cause -> Effect -> Consequence, joined by so, which means, because, therefore.
- The key link in forces is the **resultant force**: balanced (zero) means no change in speed; unbalanced means speed or direction changes.
- Weight = mass × gravitational field strength (use g = 10 N/kg today).

Q1 (1 mark) Calculate the weight of a 6 kg mass on Earth.

Q2 (3 marks) A box is pushed forwards with a force of 40 N. Friction of 15 N acts against it. (i) State the resultant force and its direction. (ii) What happens to the speed of the box? (iii) Give one reason for your answer to (ii).

Q3 (4 marks) A cyclist stops pedalling on a flat road and gradually slows down until they stop. Explain why.

Q4 (6 marks) A lorry travels at a steady 20 m/s on a flat motorway. The driver presses the accelerator harder. After a while the lorry settles at a new, higher steady speed. Explain, in terms of forces, why the lorry first speeds up and then stops speeding up.

Plan your chain first (a short phrase in each box):

Cause	Effect	Consequence	Next link

Self-check: [ ] Used the words resultant force [ ] Every sentence links to the next [ ] Units included (N) [ ] Ended by saying what happens to speed

Mark your writing: circle the level you reached.

Level	What it looks like
Level 1 (1-2 marks)	Points stated but not linked.
Level 2 (3-4 marks)	Attempts at logical linking, with gaps.
Level 3 (5-6 marks)	Points detailed and logically linked into a clear account.